Some Well-Known Part-Of-Speech (POS) Tagging Systems: A Short Survey

Niladri Sekhar Dash
Linguistic Research Unit, Indian Statistical Institute, Kolkata, India

Abstract

In natural language processing, POS tagging is a process of assigning parts-of-speech to words that are used in a text. It tries to capture some linguistic properties and functions of words used in a corpus. It is a complex process embedded with several theoretical and technical issues relating to identifying words and determining their lexicosemantic identity and syntactico-grammatical roles in a text. It also involves a process of defining the basic hierarchical modalities of tag assignment and designing rule-based schemas that are applied for the automatic assignment of tags to words. It uses a strategy with a combination of linguistic and extralinguistic knowledge and computation to achieve success. The output is a POS-tagged corpus which is a useful resource for language processing, language computation, machine learning, cognitive processing, data mining, information extraction, language teaching, dictionary compilation, and language description. Keeping these issues in view, I attempt to briefly describe a few POS tagging systems that are widely used across all major languages in this paper. This paper is for those scholars who come from linguistics and want to explore areas of corpus linguistics and language technology with a mission to serve their mother languages.

Keywords: annotation, tagging, part-of-speech, morphology, syntax, semantics, context

1. Introduction

In natural language processing, language technology, and computational linguistics, POS tagging involves the assignment of particular parts-of-speech to words in a text. Due to its diverse functional roles, it is also known as grammatical annotation (GA) and word category disambiguation (WCD). The primary tasks involve understanding the grammatical roles of words, marking each word with a part-of-speech, and disambiguating words at the grammatical level (Søgaard, 2010). The task of assigning POS value is based on the form and function of a word in the context of its use in larger syntactic frames (e.g., *phrases, sentences, paragraphs*). Although the process of POS tagging is complicated and error-prone, it is essential because research and development works of language technology cannot move further with a list of words obtained from a corpus without any information about their grammatical behaviours in various usage-based contexts. In descriptive and applied linguistics, a corpus tagged at the POS level is a useful input for both theoretical and

applicational purposes (Manning, 2011). A POS-tagged corpus presents words that are marked with different parts-of-speech in different textual contexts. It provides scopes for revising existing knowledge, traditional grammar, and dictionaries.

The fact that words vary in part-of-speech is not a new thing to natural languages. This is noted and discussed in grammar and dictionaries. However, these traditional knowledge texts hardly define the contextual conditions and factors that trigger a change in the part-of-speech of words. This question is partly addressed through POS tagging as it provides positional and contextual information to know how and why words, when used in a text, change in part-of-speech. It also shows that there are words in a language, which not only vary in part-of-speech but also in form and meaning to be recognized as 'ambiguous' (covering both *homonymy* and *polysemy*) forms. This feature is prevalent in all natural languages. For instance, in English, the word *sound* is tagged with different parts-of-speech based on the context of its use the following examples show.

(1) The *sound*$_{[NN]}$ of this music is very good.
(2) He has taken a *sound*$_{[ADJ]}$ decision.
(3) In this context, she *sounds*$_{[FV]}$ rational.

At the time of POS tagging of the words stated above, one indicates that the word *sound* is a noun (NN) in the 1st sentence, an adjective (ADJ) in the 2nd sentence, and a finite verb (FV) in the 3rd sentence. A native speaker identifies these words as a noun, adjective, or finite verb, respectively, to perform grammatical and semantic analysis based on his internalized linguistic rules and grammar. A computer, which is designed to tag words automatically, has to be programmed with linguistic rules and conditions to perform the same task of identifying parts-of-speech of words.

It is argued that the parts-of-speech that are found in grammars and dictionaries are enough for a language to tag words found in texts. In reality, it is found that there are some lexical categories and sub-categories which should be distinctly represented if a POS tagging scheme is to be designed for a language. For instance, in Bengali, there are eight parts-of-speech, namely, *noun, pronoun, adjective, adverb, finite verb, non-finite verb, postposition, and indeclinable* (Chattopadhyay 1995). It is said that only these parts-of-speech are enough for Bengali; once these are learned, there is no problem in identifying part-of-speech of words used in Bengali texts. However, after analyzing a Bengali corpus it is found that there are some more parts-of-speech (e.g., *demonstratives, infinitives, conjunctions, enclitics, quantifiers, particles*) which are not found in grammars and dictionaries. This additional set needs to be analyzed and described before one designs a POS tagging system for Bengali.

Before I move further, I need to address some issues regarding the presentation of information in this paper. The first issue is related to the justification of the paper. Why this

short paper is written can be understood if the present situation of language technology in India is visualized with reference to its application in industry and its impact on the new generation of scholars. Since the idea of POS tagging is not a very well-known domain of academic exercise, there is a need to introduce this to the readers. This paper tries to serve this purpose. It also tries to simplify the complex ideas and processes of POS tagging to general readers so that readers can comprehend how this new area is going to help them in the analysis and utilization of their languages in various domains of their linguistic activities. The target readers of this paper are not advanced scientists who are involved in the computational analysis of language data and utilizing the same for developing various language applications; the target readers are students and young scholars of linguistics who are trying to delve into various advanced domains of language processing and machine learning. This paper can be useful for them in various ways; they will understand the basic concepts, goals, and stages of POS annotation as well as know about some of the best-known and widely used POS tagging methods that are used across languages. This paper will provide necessary insights to the readers to learn about POS tagging in detail to develop similar processes for their languages.

2. What is Part-of-Speech (POS) Tagging?

Part-of-speech (POS) tagging is a process of assigning part-of-speech tags to each word used in a text after the word passes through morphological and grammatical analysis (Garside 1995). A set of designed codes (known as 'tags') carrying grammatical information are assigned to words to indicate their parts-of-speech. In most cases, a set of linguistic rules are applied to identify and assign POS tags to words to determine their meaning and grammatical functions (Bada, *et al.*, 2012). The advantage is realized at 3 levels (a) at the orthography level it distinguishes homographic words in a text to make distinctions in semantic information of words, (b) at the word level it allows to analyze of the morphological structure of words that represent their surface forms, and (c) at sentence level it allows to identify the syntactico-grammatical functions of words to assign their POS identity. It is a common form of text annotation and is considered the first stage of a more comprehensive process where multiword expressions (e.g., *phrases, compounds, reduplications, idioms, and proverbs*) are marked with special tags leading to the further assignment of phrase markers to sentences. Although the use of POS tags on words in a text makes a text cumbersome, difficult to read, and hard to comprehend by human beings, it is quite useful for providing processed linguistic data and information needed by a system for differentiating between the words different is parts-of-speech in a text (Leech and Eyes 1993). From the application point of view, POS tagging increases specificity in data retrieval from a text and provides basic grammatical information of words needed in sense tagging, morphological analysis parsing, dictionary-making, grammar writing, language teaching, and description. The process of POS tagging follows the following eight steps:

(a) Identification of words within a piece of text,
(b) Identification of their orthographic forms and appearances,
(c) Analysis of their morphological structures and formation,
(d) Identification of their syntactic (grammatical) functions in a sentence,
(e) Determination of their grammatical roles and parts-of-speech,
(f) Identification of their semantic roles in the sentences,
(g) Assignment of POS tags—either manually or automatically, and
(h) Final verification and validation of the tags assigned to words.

The POS tagging is carried out on a text at three stages: (a) Pre-editing of a corpus, (b) POS tagging to words, and (c) Post-editing of a tagged corpus (Fig. 1).

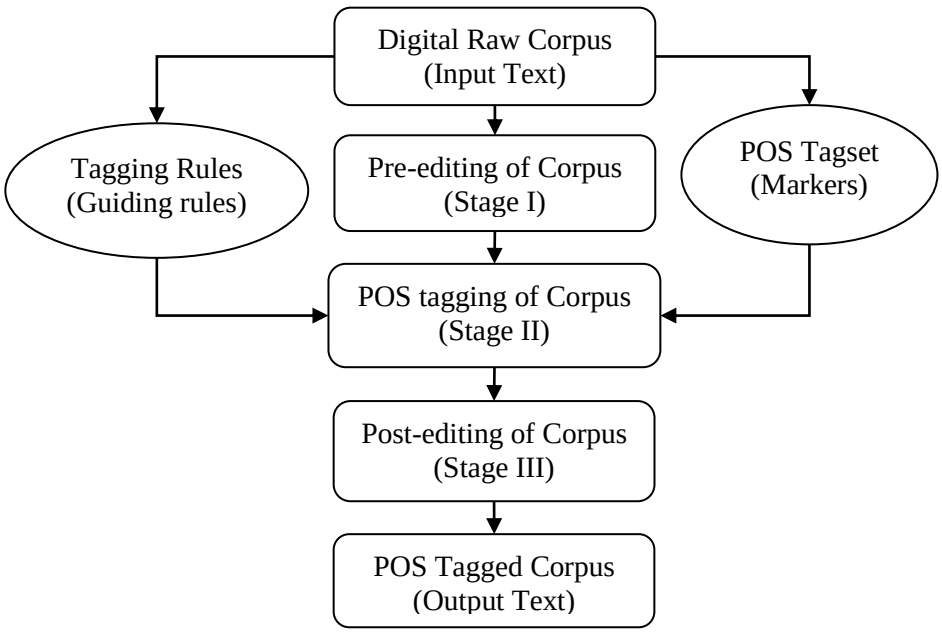

Fig. 1: Different stages of POS tagging of a corpus

We need good knowledge of the morphology and grammar of a language to do POS tagging. If a system is to do this work, it has to be trained with data, information, and rules to do the work. That means the tool, which is designed for automatic POS tagging, should be programmed with elaborate linguistic knowledge and information. Only then it can assign the correct part-of-speech to words in a text (Kupiec 1992). Moreover, before the task of POS tagging is executed on a written text corpus—manually or automatically—there is a need for a well-defined POS tagset, which will be used in a uniform manner in POS tagging. Keeping this rudimentary information in mind, in the next four sections, I present short discussions on the four most popular POS tagging systems that are widely used across languages.

3. CLAWS POS Tagging

The CLAWS (Constituent Likelihood Automatic Word-tagging System) tagging system is developed at *Lancaster University*, UK for tagging words in an English corpus. Over the years, it is revised several times (Garside, 1987; Leech, Garside, and Bryant, 1994; Garside, 1996; Fligelstone, Rayson, and Smith, 1996; Garside and Smith, 1997; Fligelstone, Pacey, and Rayson 1997). The latest version is used to tag the *British National Corpus* with a very high rate of accuracy (98%). Even on a raw corpus, it achieves more than 95% accuracy although the level of accuracy varies based on the type of English text. It also works as a post-processor of a tagging process. The use of a 'template tagger' improves tagging accuracy in a corpus.

The CLAWS tagsets are modified several times to make them appropriate for modern English texts. The CLAWS-1 tagset included 132 basic tags, many of which were identical in form and application to those used in the *Brown Corpus*. The CLAWS-2 tagset was revised and enlarged to include 166 tags. However, since such an elaborate tagset created problems in tagging, the number was reduced for general purposes. Thus the tagset, which was used for the *British National Corpus*, contained only 60 tags. It was designed mainly for handling larger corpora than small corpora of specialized works. The CLAWS-7 tagset was revised with the addition of tags for punctuation marks. It was further revised to produce CLAWS-8 to make finer distinctions in determiners, pronouns, and auxiliary verbs. Given below is an English text that is tagged using CLAWS-7 through an online tagging portal (Table 1).

Source Site	Michael Martin, Tue, 6 July 2021, 11:34 pm https://in.style.yahoo.com/)
Sample Text	A very common early sign of diabetes, increased thirst happens because diabetes causes sugar (glucose) to build up in the bloodstream. Normally, the kidneys process glucose, but when they become overwhelmed, the excess glucose is flushed out with your urine. Water from other body tissues is pulled along with it, leaving you dehydrated and craving fluids to replace what you've lost.
Tagging Tagset Output	Text with CLAWS POS Annotation Tagset: c7(http://ucrel-api.lancaster.ac.uk/cgi-bin/claws74.pl Horizontal
output	A_AT1 very_RG common_JJ early_JJ sign_NN1 of_IO diabetes_NN1 ,_, increased_JJ thirst_NN1 happens_VVZ because_CS diabetes_NN1 causes_VVZ sugar_NN1 (_(glucose_NN1)_) to_TO build_VVI up_RP in_II the_AT bloodstream_NN1 ._. Normally_RR ,_, the_AT kidneys_NN2 process_VV0 glucose_NN1 ,_, but_CCB when_CS they_PPHS2 become_VV0 overwhelmed_JJ ,_, the_AT excess_JJ glucose_NN1 is_VBZ flushed_VVN out_RP with_IW your_APPGE urine_NN1 ._. Water_NN1 from_II other_JJ

	body_NN1 tissues_NN2 is_VBZ pulled_VVN along_II21 with_II22 it_PPH1 ,_, leaving_VVG you_PPY dehydrated_VVD and_CC craving_VVG fluids_NN2 to_TO replace_VVI what_DDQ you_PPY 've_VH0 lost_VVN ._.

Table 1: Sample English text POS tagged with CLAWS-7 tagset

Several strategies are defined in CLAWS-7 to decide how to draw a line of distinction between correct and incorrect POS tags. This is essential as there is a need for detailed guidelines to be used in tagging practice and therefore it is created to remove confusion about what is ‘correct’ or ‘accurate’ tagging. The tagging guidelines of CLAWS-7 are useful for Indian language corpora as they contain some rules which are language-independent and relevant to all natural languages. Some modifications of these guidelines may be considered based on the requirement of Indian languages if we use these guidelines for POS tagging of Indian language corpora.

4. HMM-Based POS Tagging

The HMM (Hidden Markov Model) is a statistical method of POS tagging. It describes the sequence of potential events in which the probability of a POS is dependent only on the state which is attained in the previous event. It is based on the Markov assumption that predicts the probability of a sequence. In the case of POS tagging, it works in the following manner: If we have a sentence like *"We like green tea"* and we need to assign POS tags to each word in the sentence following our grammatical knowledge, then the possible POS tag for each word will be like the followings: 'We' (PN), 'like' (FV), 'green' (ADJ), and 'tea' (NN), respectively. To calculate the probabilities associated with the word tags, a system has to learn how likely it is for a noun to be followed by a verb, an adjective, and finally a noun. These probabilities are typically called 'transition probabilities' that connect the change from one state to the next in the system. It also needs to learn how likely it is that *‘We'* is a pronoun, *'like'* is a verb, *'green'* is an adjective and *'tea'* is a noun. These probabilities are called 'emission probabilities' that quantify the possibility of making a particular POS given in a defined sentence.

From the 1980s, researchers started using HMM to tag and disambiguate words in the *LOB* (*Lancaster-Oslo-Bergen) Corpus* (Kupiec, 1992). They used HMM as it provided scope for observing patterns of distribution of words of various POS for developing tables regarding probabilities of sequences of words as well as specifying patterns of use of words in the formation of English sentences. For instance, if the article ‘the’ occurs at a certain place within a sentence, then the probability of POS of the next word as a noun (40%) is the same as that of an adjective (40%) followed by words of other POS (20%). Based on this, they developed a program that could calculate that the word ‘cook’ in a sentence like "The cook

has left for home" is more likely to be a noun than a verb (Leech, *et al.*, 1983). It also helped to gather information about the POS of a word that followed after a noun in a sentence. The HMM could learn probabilities not only of word pairs but also of larger sequences made with three or more words (Belmore, 1991). For instance, if a tagging program first encounters an article followed by a gerundial noun (e.g., the[DET] making[NN] of[PP] a[DET] film[NN] is[FV] a[DET] long[ADJ] process[NN]), then the probability of the next word to be a preposition is much higher than others. This was useful for the development of an automatic POS tagging system for English.

The application of HMM showed that, in those situations where ambiguities occur, the probability of identifying the POS of a word is multiplied. With higher-order HMM, one can enumerate every possible combination of words and assign them a relative probability by multiplying the probabilities of each choice in a turn sequence. The highest probability is then selected as the most suitable candidate for POS tagging. This technique achieves higher accuracy in tagging on English corpora. There are criticisms against HMM arguing that in the process of dissolving ambiguities, it assigns the most common tag to each known word and 'proper noun' to all unknown words. It is not a worthy process as any POS tagger can achieve some level of accuracy in this method because most of the words are unambiguous in a text (Charniak 1997). Another limitation of HMM is noted in its 'generous nature' in tag assignment. Although HMM is used due to its high rate of accuracy in comparison to CLAWS, it is quite expensive as it tends to enumerate all possibilities. Moreover, it resorts to backup methods when there are too many possibilities creating problems in disambiguation (Atwell, 2008). This can make a POS-tagged corpus more deceptive and less reliable than it was before it is tagged.

5. Brill POS Tagging

The Brill POS tagging method is developed by Eric Brill in 1993. It generates an interface during POS tagging following pre-defined rule-based strategies (Brill 1992). It is known as 'Transformation-Based Error-Driven Tagging' that follows an inductive method and is based on the following two principles (Brill 1995):

(a) It operates through 'supervised learning' to minimize errors in POS tagging; and,
(b) It follows a transformation-based process, in the sense that a tag is assigned to each word and changed using a set of predefined rules.

In the tagging process, if a word is known to the system, the system assigns the known tag to it. If the word is not known, it assigns the tag 'noun' to it. By applying this method over-and-over the rule and changing the incorrect tags at subsequent operations, the system achieves a high rate of accuracy. It ensures that lexical information (e.g., *morphosyntactic information of*

words) is properly employed in the tagging process. The algorithms may be summarized in the following seven stages:

(1) Start the POS tagger on a digital corpus text and encounter a word.
(2) If the word is found in the inbuilt lexicon, assign the most frequent tag associated with it.
(3) Encounter a new word (not in the inbuilt lexicon) and tag it as a proper noun (if capitalized) or as a simple noun (if not capitalized).
(4) Learn or guess possible tags of the word based on contextual rules.
(5) Change the incorrect tag to the correct one with contextual rules.
(6) Finalize the POS tagging assignment to the word.
(7) Generate the final output.

The learning phase of the Brill POS tagging system involves several sub-stages and strategies. First, it iteratively computes the error score of each candidate rule to calculate the difference between the number of errors before and after applying the rule. Second, it selects the best (higher score) rule. Third, it adds it to the ruleset and applies it to the text again. Fourth, it repeats the process until no rule has a score above a given threshold. If the chosen threshold is zero, it continues the application of rules until it achieves a greater score than the chosen threshold. Once it is achieved, it is then considered to be the final stage of tagging (Fig. 2).

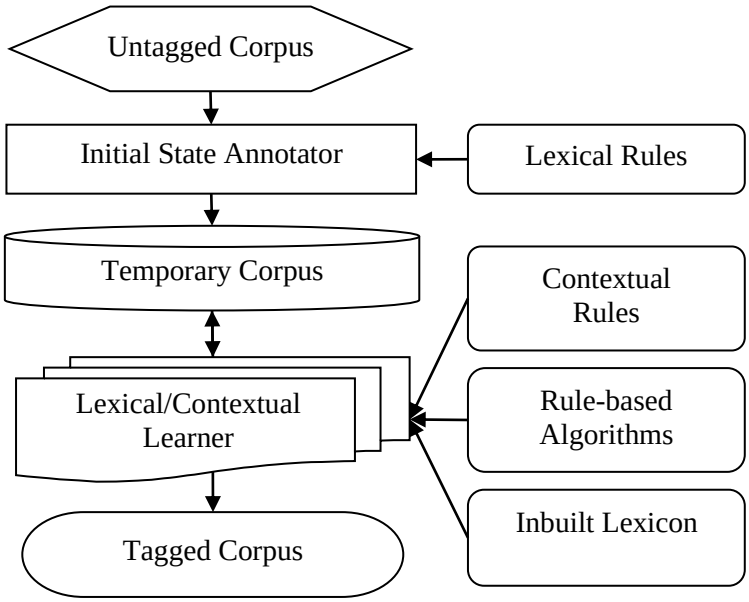

Fig. 2: The rule-based Brill POS tagging system

For achieving a higher rate of accuracy it applies two sets of rules: the first set of rules are called the 'Lexical Rules', which are used for the initialization of a POS tagging process. The

second set of rules is known as ‘Contextual Rules’, which are applied to remove errors and correct final tags.

(a) Lexical Rule: Tag W_1 → W_N IF W_1 carries suffixes ‘-tion’, ‘-ment’, etc.
(b) Contextual Rule Tag W_1 → W_2 IF the preceding or the following tag is X.

The Brill system faces problems in assigning POS to words belonging to ‘open class’ because words of this class often carry complex morphological structures. On the other hand, lexical categories of words of ‘closed class’ are easier to detect and annotate. It also shows that it is more difficult to derive information from those words which are tagged with POS information only than those words which are tagged with additional information with regard to their inflectional types (Toutanova, *et al.*, 2003). Therefore, for better evaluation of the system, it is necessary to test the Brill system on corpora of diverse text types where words are used with inflections. Higher accuracy can be achieved by using it on corpora made with different text types. Also, the use of a larger lexicon can reduce the number of unknown words. The accuracy of the system can be improved if rule-generating mechanisms are flexible allowing consideration of different characteristics of languages. Given below is a sample English text and its POS-tagged version generated through the Brill POS tagging system (Table 2).

Input Text	So many people have diabetes-about 1.5 million are diagnosed in the United States each year, and nearly 1 in 10 Americans have it-you'd think it'd be easy to spot. But although the condition is relatively common, many people go undiagnosed because the early symptoms can be vague, easily overlooked at first, or confused with other conditions.
Output Text	So/RB many/JJ people/NNS have/VBP diabetes-about/JJ 1.5/CD million/CD are/VBP diagnosed/VBN in/IN the/DT United/NNP States/NNPS each/DT year/NN ,/, and/CC nearly/RB 1/CD in/IN 10/CD Americans/NNPS have/VBP it-you/NN 'd/MD think/VB it/PRP 'd/MD be/VB easy/JJ to/TO spot/VB ./. But/CC although/IN the/DT condition/NN is/VBZ relatively/RB common/JJ ,/, many/JJ people/NNS go/VBP undiagnosed/JJ because/IN the/DT early/JJ symptoms/NNS can/MD be/VB vague/JJ ,/, easily/RB overlooked/VBN at/IN first/JJ ,/, or/CC confused/VBN with/IN other/JJ conditions/NNS ./.

Table 2: Output obtained from Brill POS tagging system

Although it is claimed that the Brill POS tagging system is language-independent (Brill 1995), in reality, it has shortcomings with those languages which are different from English.

This system is yet to be tested with Indian language texts to see how it works for these languages.

6. TnT POS Tagging

The TnT (Trigrams-n-Tagging) POS tagging is also a statistical method that is trained for different languages. It can be adopted with any tagset of a language (Brants 2000). The performance of this system depends largely on its ability in generating components for parameters through its trial use on pre-tagged corpora. Moreover, it is capable of incorporating several methods of smoothing to handle those words that are not previously encountered by the system (Ramshaw and Marcus 1995). Since this tagging system is not optimized for a particular language or a variety, one can use it for training on corpora of different languages. Also, one can apply this method to a text of a new language as well as to a new genre of a text. It can be optimized to speed up tagging so that it generates quick outputs from a corpus. It tags words after analyzing the suffix parts tagged with words. In this way, all known and unknown words are handled by a suffix trie-structure and successive abstractions. It is applied to corpora by using three different modes:

(a) Input Mode: The input file contains one token per line.
(b) Base Mode: Tagger adds the second column to each line, containing a tag for the word.
(c) Output Mode: Tagger emits alternative tags for each token with a probability distribution.

If a word is marked with an asterisk (*) in the output database, it is assumed that the word is not in the lexicon used by the system. The speed of tagging largely depends on the rate of average ambiguity of words and the percentage of unknown words used in a text. When it is applied to small corpora, it generates low accuracy (around 94%) due to the small size of corpora and the large size of the tagset (160+ plus multi-token tags). But when it is used on large corpora with reduced tagset, the percentage of accuracy is very high (around 98%). The method is trained on corpora where written words are separated by a white space (Brants 2000). A notable limitation of the TnT tagging system is that it works well with a tagset represented in the ASCII (*American Standard Code for Information Interchange*), but falters with a tagset represented in other encoding systems.

7. A Brief Note on the Indian Scenario

Perhaps the first POS-tagged text for any Indian language is developed in 2004 before any attempt is made to design a tool for this. We manually POS tagged a Bengali text of a

hundred thousand words to calculate the frequency of use of words of different POS in Bengali (Dash 2004). We used a POS set of standard Bengali grammar to form the first POS tagset for Bengali (Table 3).

No.	POS Categories	Tag	Example
1	Noun	[NN]	bālak, śahar, kathā, mānuṣ
2	Pronoun	[PRN]	āmi, tumi, se, tārā, tui
3	Demonstrative	[DMS]	ýe, ei, oi, tāi
4	Finite Verb	[FV]	karchi, kartām, gela, ýābe
5	Non-Finite Verb	[NFV]	karle, karte, gele, giye
6	Adjective	[ADJ]	bhāla, manda, sundar,sādā
7	Adverb	[ADV]	haṭhāt, bābad, kāraṇe
8	Postposition	[PSP]	pare, kāche, āge, niche
9	Conjunction	[CNJ]	tabe, ýadi, naile, ýāte
10	Indeclinable	[IND]	kintu, athabā, baraṃ ār
11	Particle	[PRT]	i, o, to, nā, ne, ni
12	Quantifier	[QNT]	ek, dui, pratham, paylā
13	Punctuation	[PNC]	., : ; - / …, !, ? ( ), [ ], { }, etc.
14	Symbols	[ORS]	+, -, x, >, <, $, #, @, ^, &, * etc.

Table 3: Generic POS tagset proposed for Bengali (Dash 2004)

The work of POS tagging was carried out manually on a small Bengali text corpus (Dash 2005: 104-108). As shown below (Fig. 3), there was no attempt to go into sub-classification of POS as we found in English (Biber *et al.* 1998, Barnbrook 1998). Since it was the first attempt, we tried to provide only broad POS level information to words based on their role in texts.

<Sentence START>
kṛṣi[NN] kārýer[NN] janya[PP] ek[ADJ] jāygāy[NN] saṅgabaddha[ADJ]
haiyā[NFV] basabās[NN] karite[NFV] lāgila[FV] ebaŋ[IND] parasparer[PN]
madhye[PP] kṛṣi[NN] kārýer[NN] bibhinna[ADJ] kalākauśaler[NN] biṣaye[NN]
labdha[ADJ] jñān[NN] binimay[NN] dvārā[PP] kṛṣikei[NN] jībikār[NN]
subidhājanak[ADJ] upāy[NN] hisābe[PP] grahaṇ[NN] karibār[NFV]
siddhānta[NN] grahaṇ[NN] karila[FV] ebaŋ[IND] śikār[NN] apekṣā[PP] kṛsi[NN]
kārýer[NN] sāhāýye[PP] jībikā[NN] arjaner[NN] upar[PP] mānuṣ[NN]
adhik[ADJ] jor[NN] dila[FV] .[PNC]
<Sentence END>

Fig. 3: Sample of a POS tagged Bengali written text (Dash 2005)

In another experiment, to add more information at the word level, we attempted to provide multiple levels of POS information to words (Dash 2005: 112). Since the use of homonyms in a text is quite regular, it was necessary to identify the possibility of a word to be used in different POS in a text. Given below is an example, where some Bengali words, following Biber *et al.* (1998), were tagged with multiple POS values to inform that based on contexts a word may be used in different POS in Bengali (Fig. 4).

POS Annotated Text (Bengali)	
ābdul_NN ++++	**chini_FV/NN ++++**
mājhi_NN ++++	**se_PRN/ADJ ++++**
chhũchālo_ADJ ++++	dādāke_NN ++++
tār_PRN/NN ++++	padmā_NN ++++
dāṛi_NN ++++	**theke_PSP/NFV ++++**
gõph_NN ++++	ene_NFV ++++
tār_PRN/NN ++++	dita_FV ++++
kāmāno_ADJ/FV ++++	iliś_NN ++++
māthā_NN ++++	māch_NN ++++
tār_PRN/NN ++++	**ār_IND/ADV**
neṛā_ADJ/NN ++++	kachchhaper_NN ++++
tāke_PRN/NN ++++	ḍim_NN ++++

Fig. 4: Multiple POS tagging of words in Bengali (Dash 2005)

In the last three decades, there have been efforts to build a POS tagset and POS-tagged corpus in Indian languages. These efforts are primarily computer-oriented because POS tagging is a computationally ‘easy’ problem compared to a manual attempt. For the Indian languages, POS tagging is no more a theoretical exercise. Since I cannot give details about these tools in this paper, I refer to only a few tools that address POS tagging in Indian languages (Saha *et al.* 2004, Avinesh and Karthik 2007). Also, there are reports on POS tagging and chunking (Dandapat 2007, Rao *et al.* 2007, Sastry *et al.* 2007, Saharia 2009) and POS tagging and parsing (Rao and Yarowsky 2007). An attempt is also made to develop a common POS tagging framework for all Indian languages by using a standard POS tagset (Baskaran *et al.* 2008). Scholars have used mathematical models, statistical tools, and computational techniques to develop POS tagging tools and trained them on Indian language corpora with nearest-neighbor transition probabilities to achieve high accuracy. To date, the POS tagging tools that are developed for Indian languages, include Hindi (Shrivastava and Bhattacharyya 2008, Ray *et al.* 2010, Mishra and Mishra 2011), Bengali (Ekbal *et al.* 2007, Dandapat 2009, Chakrabarti 2011), Kannada (Shambhavi *et al.* 2012), Tamil (Dhanalakshmi *et al.* 2009), Malayalam (Antony *et al.* 2010), Assamese (Saharia *et al.* 2009), and Bhojpuri

(Singh and Jha 2015). Attention is also given to other Indian languages (Rao *et al.* 2007, Pammi and Prahallad 2007, Rao and Yarowsky 2007, Kumar and Josan 2010, Shambhavi and Ramakanth 2010). A milestone in this area is the building of a parallel POS-tagged translation corpus in 23 Indian languages (Jha 2010). This has opened up many possibilities for using POS-tagged texts in linguistics ad language technology (Dash *et al.* 2016).

8. Summary of Observations

The POS tagging methods that I have discussed in this short survey paper mostly use pre-existing digital corpora to experiment and apply methods of POS tagging algorithms. These experiments give us an important insight: it is possible to bootstrap texts by using 'unsupervised' tagging conventions. In most cases, an unsupervised tagging technique uses a small part of a tagged text corpus for training purposes, and then it produces desired outputs through induction of new rules and information either derived from the contexts (rarely) or obtained from the inbuilt lexicon provided within a system. In most cases, a system observes patterns of word use in raw texts and then derives the POS information for words from the texts after calculating possible morphosyntactic identities of words in a sentence. For example, information gathered from raw English corpora reveals that while the determiner *'an'* occurs in limited contexts, the determiner *'the'* occurs in much more varied contexts. It implies that the application of probability measurement, with sufficient iterations and repeated use in varieties of text types, can generate similarity scores for words of similar POS—a crucial strategy that linguists want to prepare for a POS tagging system. Also, the application of this technique produces differences, the analysis of which provides valuable insights about the POS of words, hardly presumed by human annotators.

Tagging a word at the POS level on a text is full of unforeseen challenges. We have to keep in mind that POS tagging—either we do it manually or automatically—we have to do it on a piece of raw text. Typically, a text contains many words, which occur in sentences with specific syntactic functions and semantic roles. It is a challenge for us as well as a machine to identify the specific POS value of a word. We cannot do this unless we understand the actual morphological-cum-syntactic role of a word in a text (Toutanova and Manning 2000). Also, we have to consider linguistic and technical issues involved in it before we start POS tagging to words. In most cases, these issues are related to the sanitation of text, correction of sentences, normalization of words, tokenization of inflected words, removal of spelling errors, validation of real words, management of white space, and correction of grammatical errors (Dash 2021b). We can avail linguistic information about a word from its structure, meaning, syntactic role, and functional relation with other words used in a sentence (Brill 1992). While structure refers to its morphological features, lexical properties refer to its abstract or concrete values, syntactic role refers to its subject-predicate role in a sentence, and semantics refers to its agentive or thematic role in a sentence. We may utilize such

information during POS tagging to ensure that the linguistic information of a word is rightly represented and marked (Durand *et al.* 1996).

Variation in the POS of a word is a common thing in all languages. Also, the number of POS mentioned in the standard grammars of a language is enough for tagging words in a language. We, however, find that a word can be used in more POSs than registered in grammar. We should identify and describe the additional POSs when we design a scheme for POS tagging. For instance, we have learned that Bengali has 8 POS: *nouns, pronouns, adjectives, adverbs, finite verbs, non-finite verbs, postpositions*, and *indeclinables*. We are taught that once we learn these POSs, we can identify the POS of a word used in a Bengali text. We are also advised that these POSs are adequate for knowing words in Bengali texts. In reality, however, we find that there are some more POSs (e.g., *quantifiers, conjunctions, demonstratives, enclitics, particles*) that deserve attention. We should analyze, understand, and define the forms and functions of these POSs to gather better ideas about words used in texts. In the actual work of POS tagging, we face more challenges when we try to distinguish between plural and singular nouns, tag words with case makers; identify nouns and adjectives with gender and number; tag verbs with a person, number, tense, aspect, modality, and honorification; tag adjectives with degree suffix; tag adjectives that behave like nouns. These are more complex and challenging problems as they are loaded with various grammatical information. If these challenges are not addressed, a POS tagging tool fails to mark unique linguistic information of words. As a result, a tagged text fails to reflect on the finer linguistic properties of words.

Despite so many advantages of POS tagging, there are limitations regarding the proper identification of POS of words used in texts. This leads scholars to argue that POS tagging should be separated from other types of corpus tagging (e.g., *semantic tagging, syntactic tagging*, etc.). It is also realized that since POS tagging, chunking, and parsing are three different ways of tagging a corpus with different goals, they should be addressed separately. This helps to systematize various layers of text tagging, separate POS tagging from other related tasks, and devise more useful methods for POS tagging. In the last fifty years, POS tagging is accepted as an indispensable exercise in corpus tagging. It has become useful for designing text-based systems and tools that are used in language-related services to common people through digital platforms. Recent studies report on some high accuracy-based POS tagging tools for English and other advanced languages. Similar success for Indian languages has not yet come, although there are some POS tagging tools, which are quite robust (Dash 2021a).

What we realize from this paper is that there are large numbers of words, the correct POS of which cannot be assigned without understanding the contexts of their occurrence and the meanings they denote in different contextual environments. We also realize that we need to understand the information underlying the structure and discourse of a text to assign the right

POS tags to words. These are complex and expensive pre-conditions for developing a useful POS tagging system because extraction of information from an extra-textual level and applying the same in a POS tagging system is a complex machine learning task. Even in a simple linguistic study, analyzing information of higher levels relating to meaning, function, pragmatics, and discourse is a complex cognitive challenge. Therefore, a system developer, who is trying to develop a POS tagging tool, has to be extremely skilled in transferring linguistic knowledge and tagging complexities into an application. A workable solution is the application of a POS tagging tool on the corpus and improving the precision of the tool with continuous analysis of outputs and modifications of POS tagging algorithms. This is still a dream for most of the Indian languages.

9. Value of a POS Tagged Corpus

The value of a POS-tagged corpus is enormous. It is used in language description, language teaching, grammar writing, dictionary compilation, text processing, and language-based technology (Zhou, *et al.*, 2005). One can have some ideas on how a POS-tagged corpus is utilized in applied linguistics, computational linguistics, and cognitive linguistics from the following diagram (Fig. 5). In language description and application, a POS-tagged corpus is used to describe the form and nature of inflection of words, record patterns of use of inflected words, calculate the frequency of use of words of different POS, define patterns of distribution of different lexical tokens, understand nature of changes words undergo during inflection, compile lexical lists of different POS, and teach processes of formation of words of different POS to language learners. In language computation, a POS-tagged corpus is used to develop systems for word sense disambiguation, named entity recognition, chunking, sentence annotation, text understanding, query answering, grammar checking, e-learning, and machine translation. It is also used in machine learning, extraction of linguistic and grammatical elements, and language modelling.

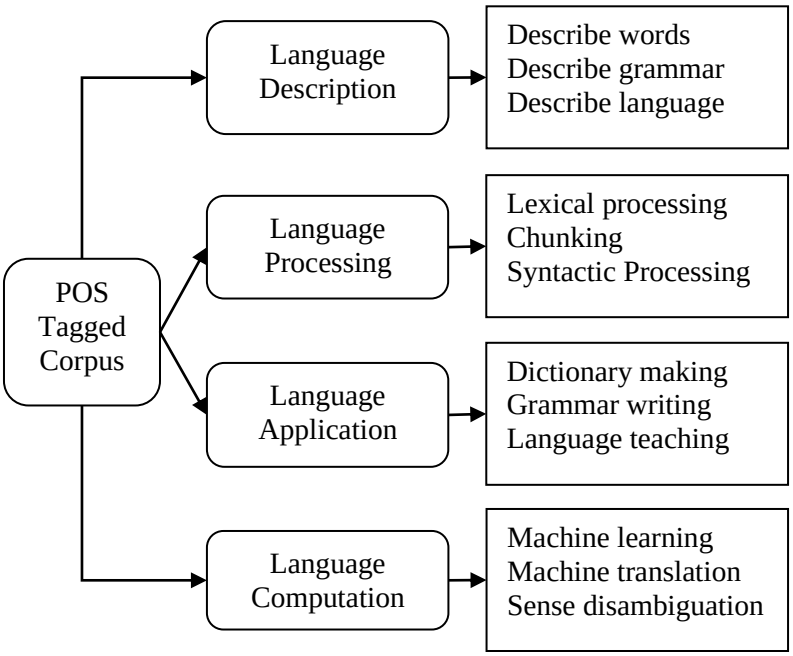

Fig. 5: Utilization of POS annotated text in various domains

Keeping the multipurpose use of a POS-tagged corpus in mind, I propose here to take initiative to develop POS tagged corpus for all the scheduled and non-scheduled Indian languages. During this work, we should keep the following issues in mind:

(a) A corpus should be pre-edited, normalized, and standardized before it is used in POS tagging,
(b) The tagset should be large keeping in view the existing POS varieties of an Indian language,
(c) The tagset should be designed to optimize accuracy in POS tagging, and
(d) The POS tagging system should work well for all types of synchronic and diachronic texts.

This is a short survey paper where I refer to a few notable POS tagging systems that are used in many advanced languages. Although many research papers have discussed different types and methods of POS tagging, I hardly come across any paper that summarizes the basic concepts and functionalities of the major POS tagging systems. From this perspective, this is a ‘gateway paper’ (very brief in length, though) in the sense that it gives some preliminary ideas about an area of study which requires deeper and wider exploration. One of the major limitations of the paper is that it fails to refer to any POS staging system that has been successfully developed and used in Indian languages. This is because, to date, no good quality POS tagging system is made openly available for public access for the Indian languages although it is known that some good quality POS tagging systems are developed and used in some Indian languages. Another limitation of the paper is that, due to paucity of space, I am not able to describe POS tagging in detail regarding actual computational

operations of a system as well as refer to some new and more advanced POS tagging methods that require advanced knowledge in computation, statistics, and linguistics. Hence, I try to delimit myself within a specified scope keeping in mind that the paper desires to serve some pedagogical purposes rather than presenting results of any ground-breaking innovation.

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